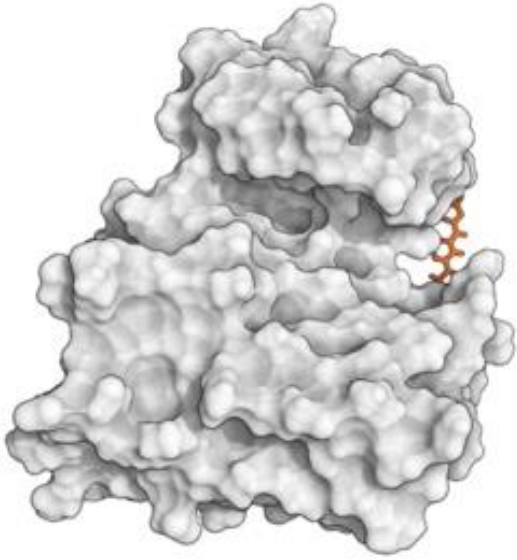
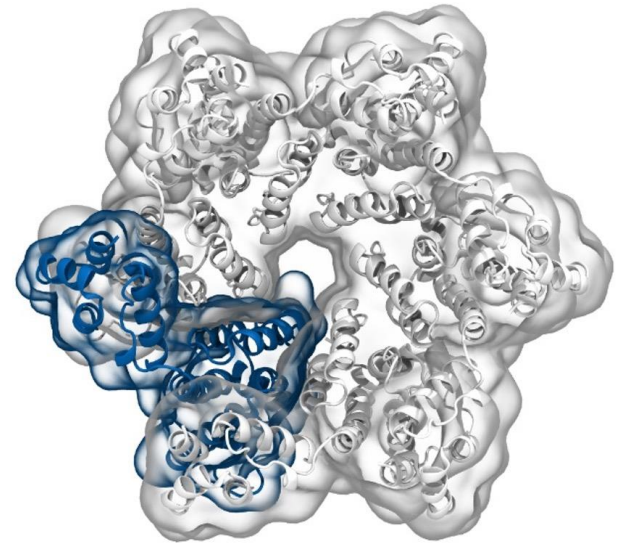


Simulation of Biomolecules



Protein preparation



Dr Jasmin Güven

University of Bristol

jasmin.guven@bristol.ac.uk

Dr Matteo Degiacomi

University of Edinburgh

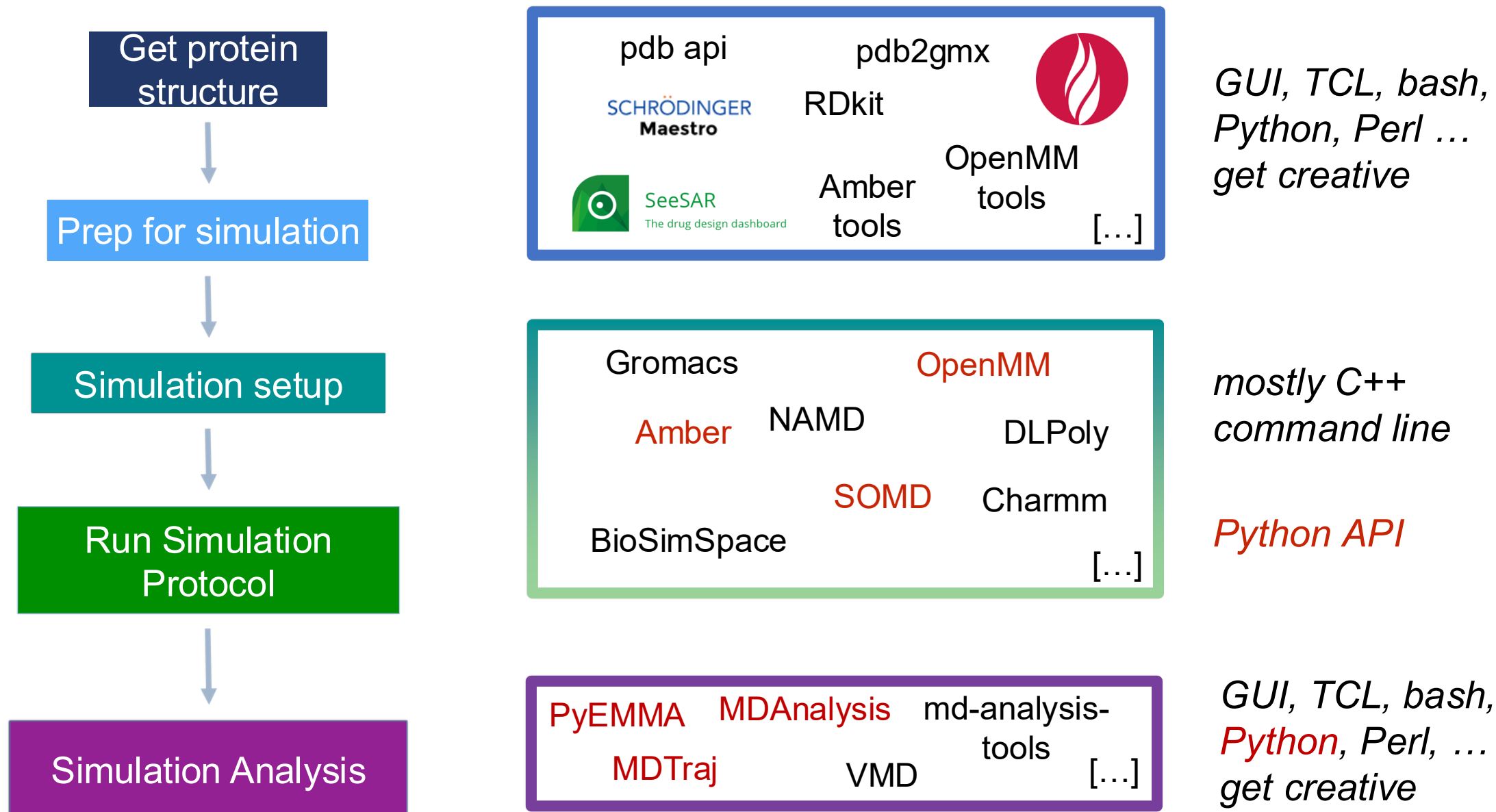
matteo.degiacomini@ed.ac.uk

Dr Antonia Mey

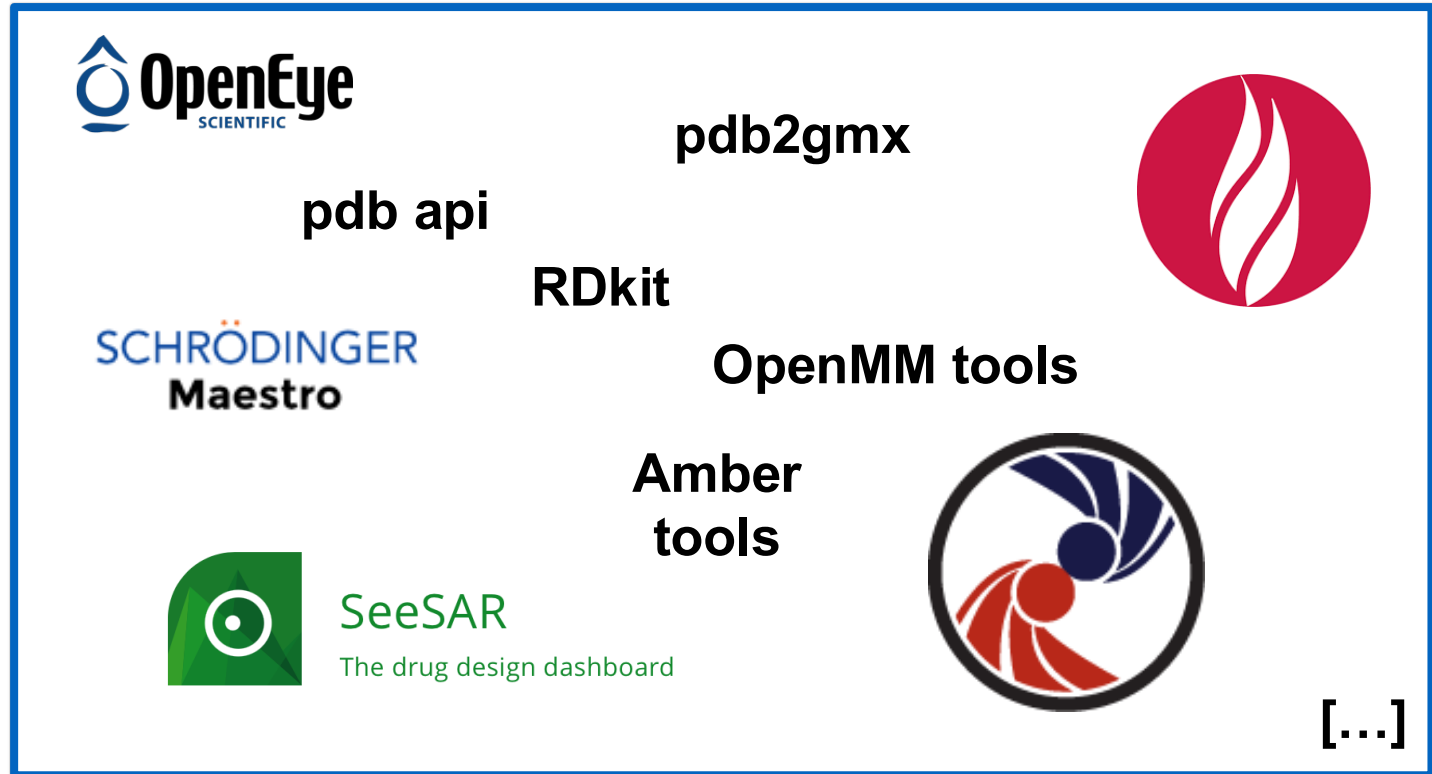
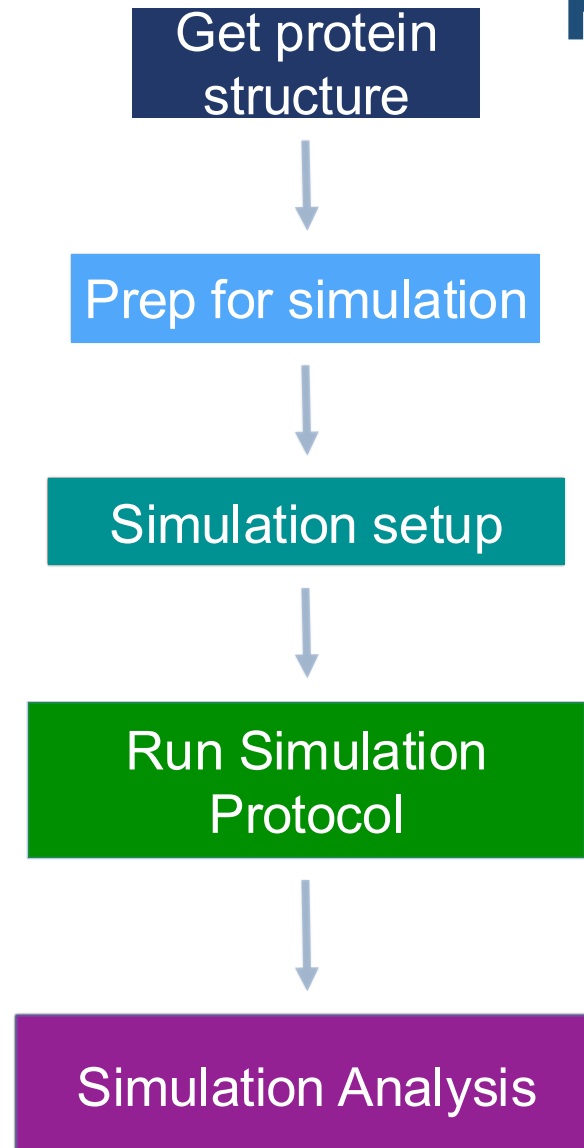
University of Edinburgh

antoniamey@ed.ac.uk

A typical workflow for molecular dynamics



Let's get started with understanding protein structures



GUI, TCL, bash, python, Perl, ... get creative

Crystal structures are models with potential errors

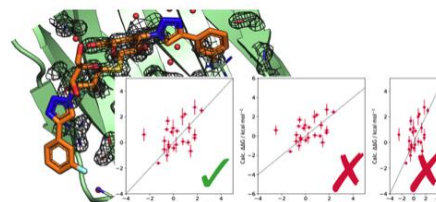
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Best Practices for Constructing, Preparing, and Evaluating Protein-Ligand Binding Affinity Benchmarks [Article v1.0]

David F. Hahn

Computational Chemistry, Janssen Research & Development, Turnhoutseweg 30,
Beerse B-2340, Belgium

<https://orcid.org/0000-0003-2830-6880>



Gert Vriend

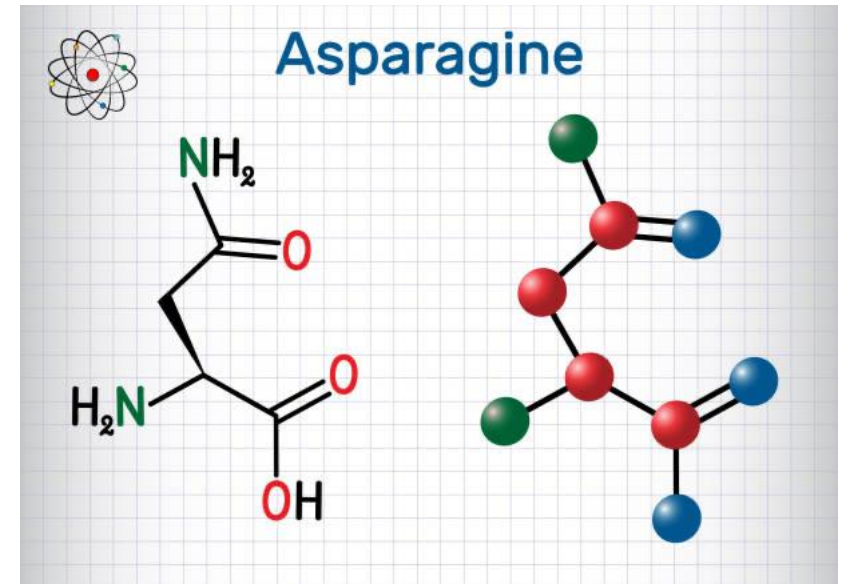
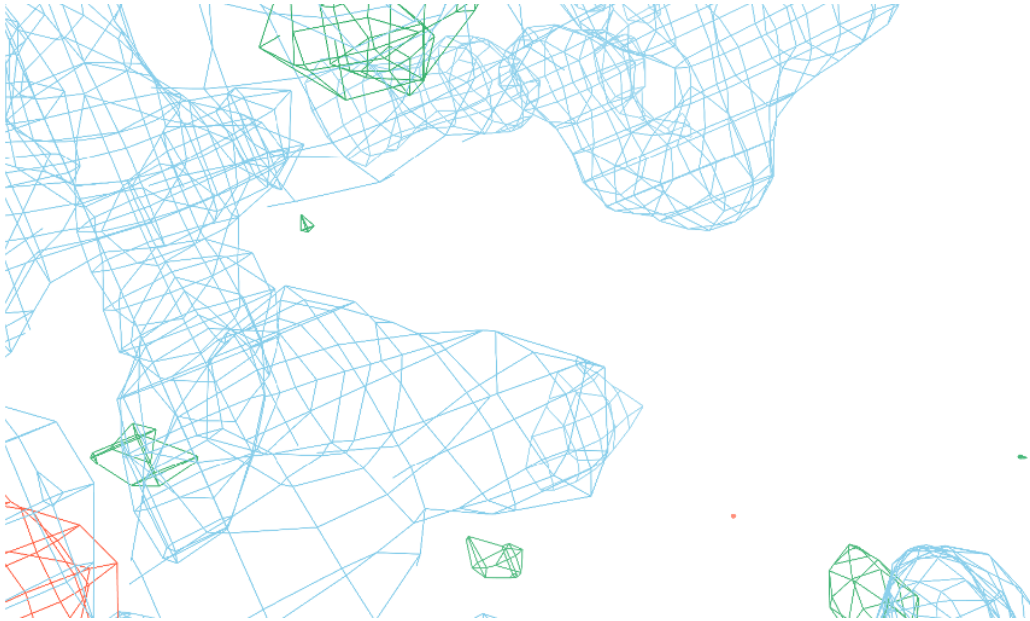
Examples of errors/oddities in PDB structures:

<https://swift.cmbi.umcn.nl/teach/pdbad/>

Crystal structures are models derived from electron densities

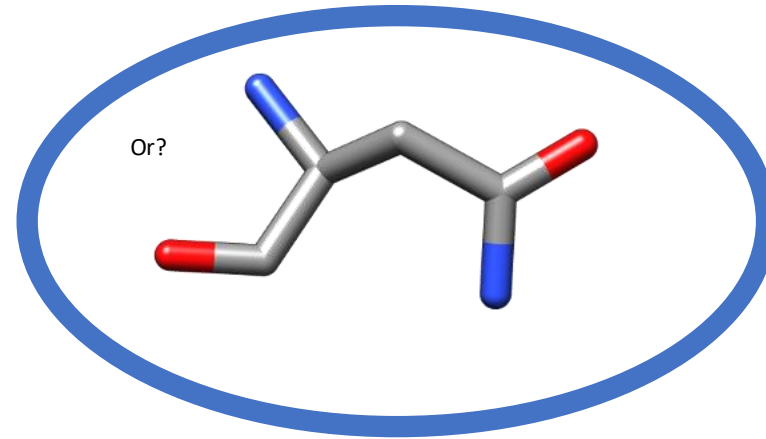
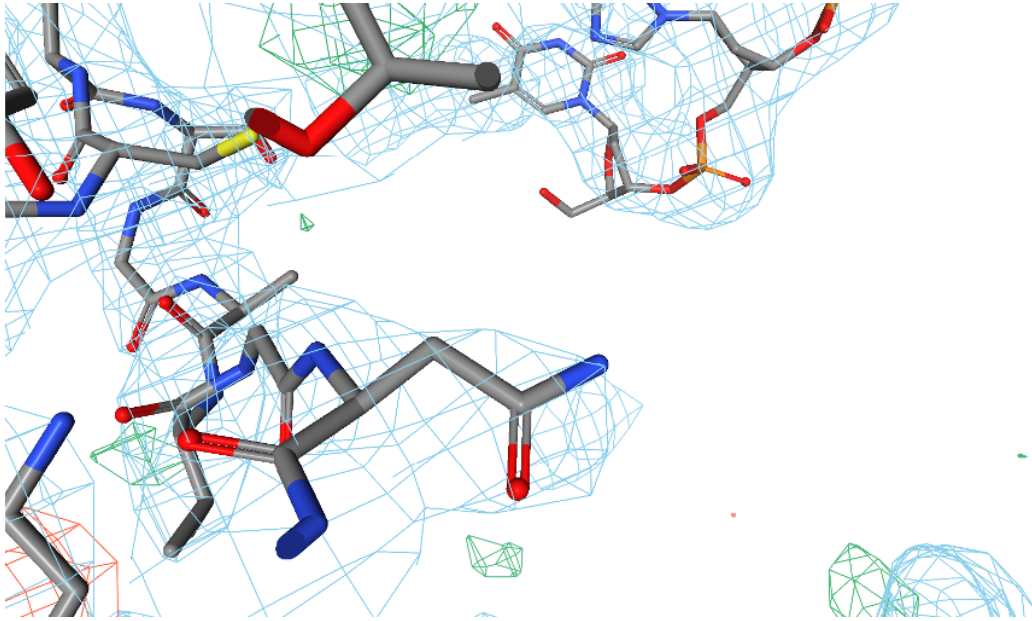
1T38: HUMAN O6-ALKYLGUANINE-DNA ALKYLTRANSFERASE

ASN157



Crystal structures are models derived from electron densities

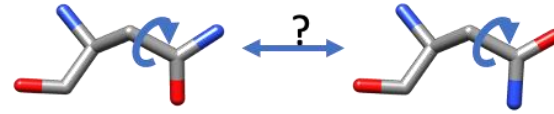
1T38: HUMAN O6-ALKYLGUANINE-DNA ALKYLTRANSFERASE



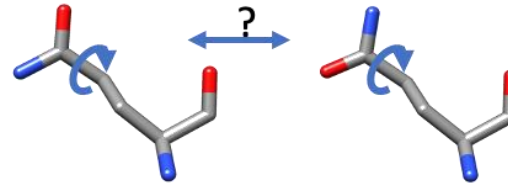
Alternative conformations of side chains - NGH flips

Typically, the crystallographers will have assigned the orientation based on potential H-bonding interactions, etc.

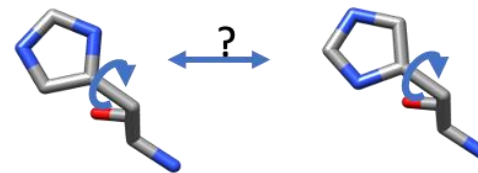
No standard protocol for this:
always worth double-checking.



Asparagine (N)



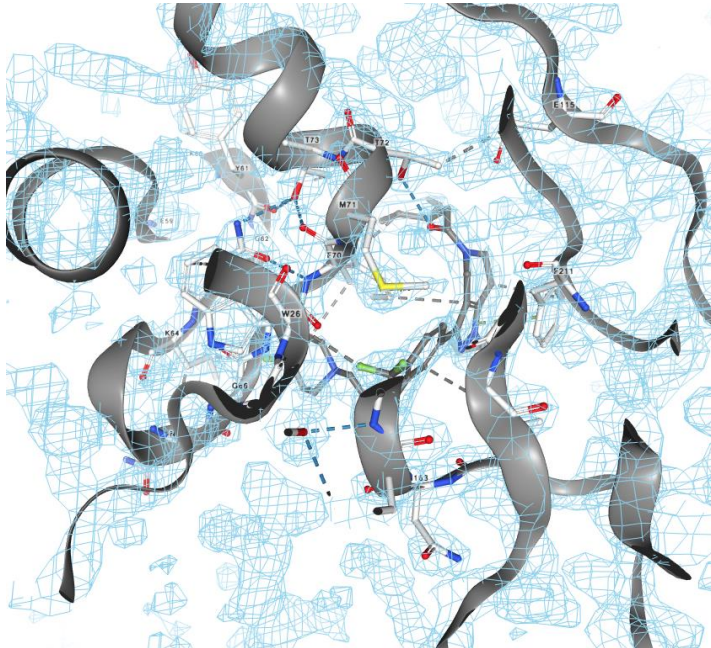
Glutamine (Q)



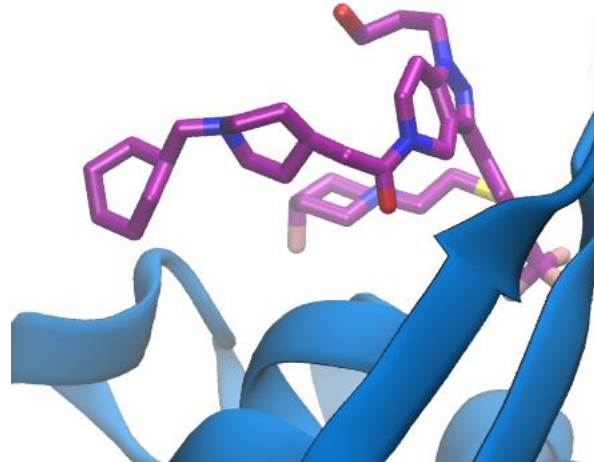
Histidine (H)

Ligand densities can also have creative input from the crystallographer

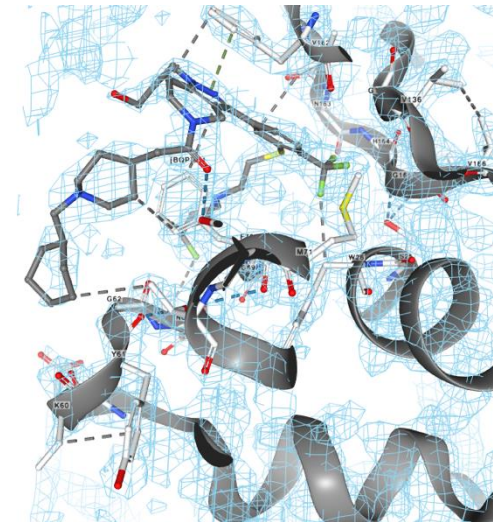
Cathepsin S



Cathepsin S
with ligand

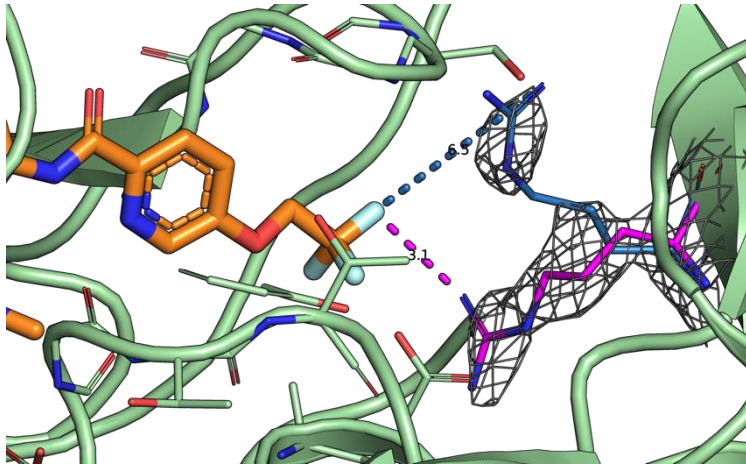


boat/chair?

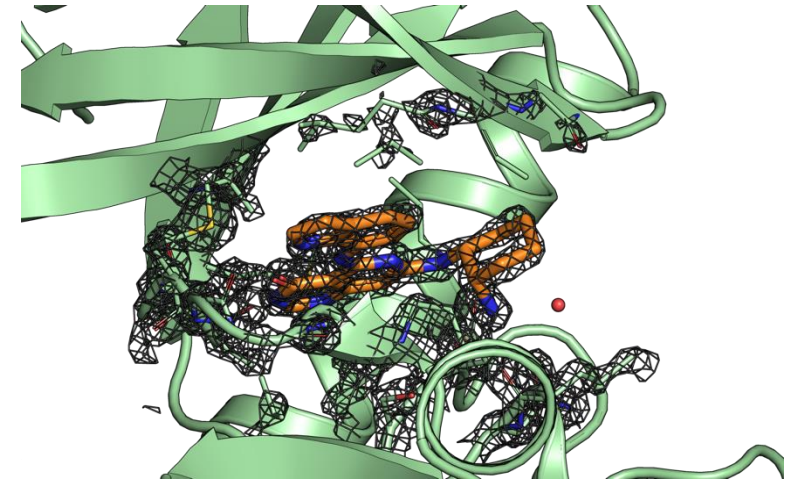
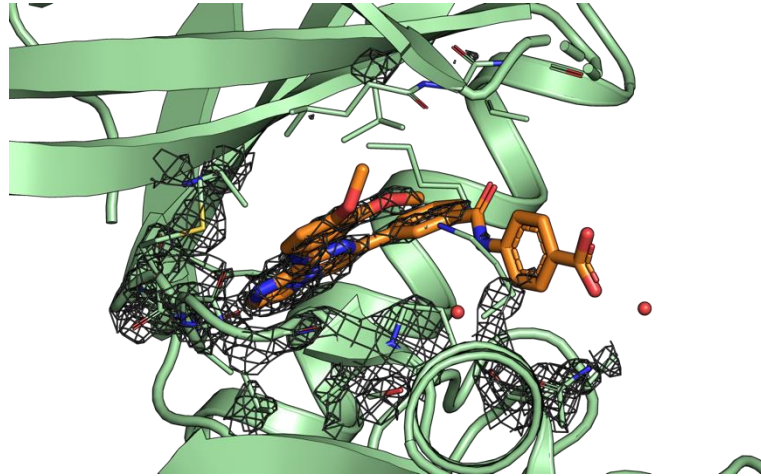


Picking the best crystal structure requires care

BACE (Hunt)



spleen tyrosine kinase

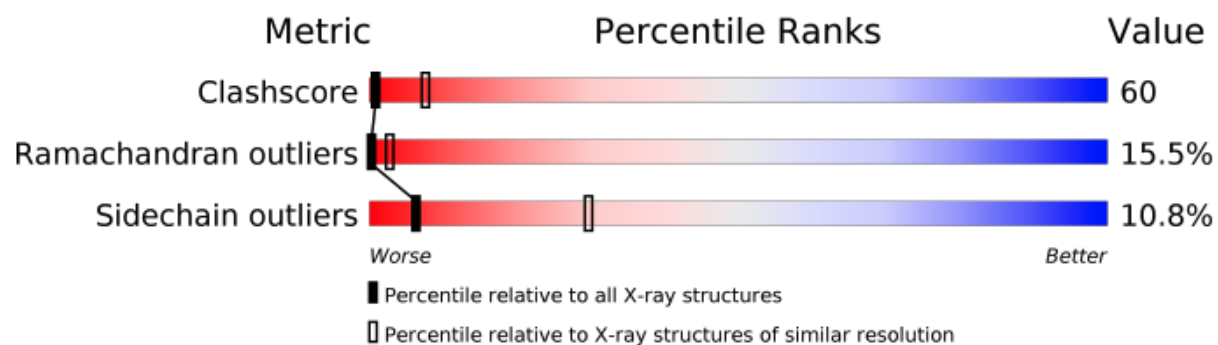


Which protein structure?

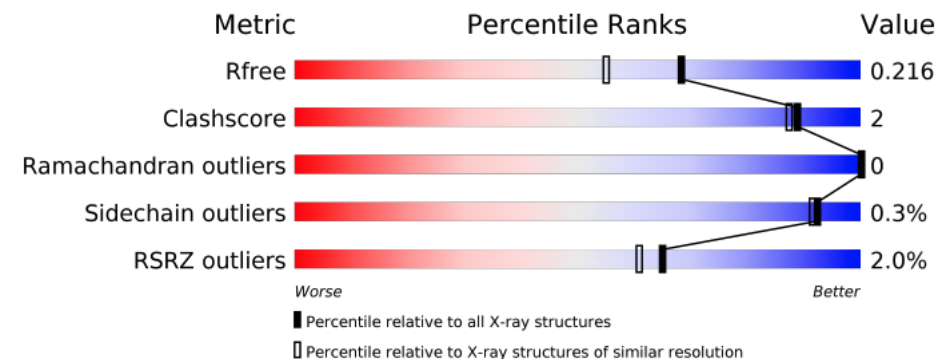
- Alternative side chain conformations need to be assessed carefully
- Active site residue densities are important for choosing the right crystal structure

The RCSB PDB report can help with choosing structures

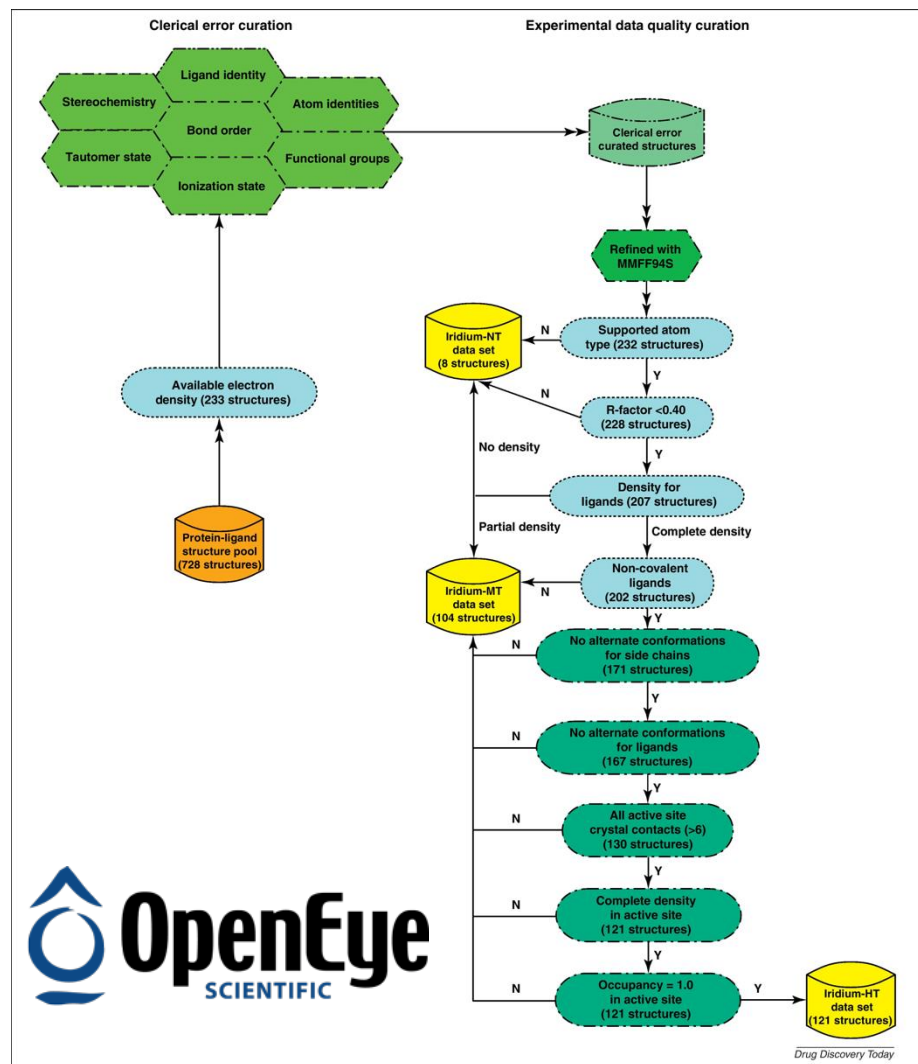
Jnk1 - 2GMX



Jnk1 - 3ELJ



The Iridium score can help assess the trustworthiness of an X-ray structure



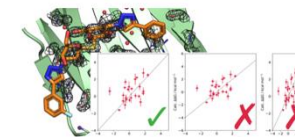
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Best Practices for Constructing, Preparing, and Evaluating Protein-Ligand Binding Affinity Benchmarks [Article v1.0]

David F. Hahn

Computational Chemistry, Janssen Research & Development, Turnhoutseweg 30, Beerse B-2340, Belgium

<https://orcid.org/0000-0003-2830-6880>



Jnk1[74, 86]
MCL1[74, 87]

2GMX (NT, -) ^f	0.77
4HW3 (HT, 0.41)	0.26

6D09 (HT, 0.35) ^b
3ELJ (MT, 0.31) ^a
6O6F (HT, 0.30)
4ZBF (HT, 0.35) ^b
3WIX (HT, 0.37) ^{a, c}

3V3V (MT, 1.5) ^{b, h, e}	21	3.4
4WMU (HT, 0.41) ^b	42	4.2
4ZBI (HT, 0.45) ^b		

12 Features to inform the Iridium Score:

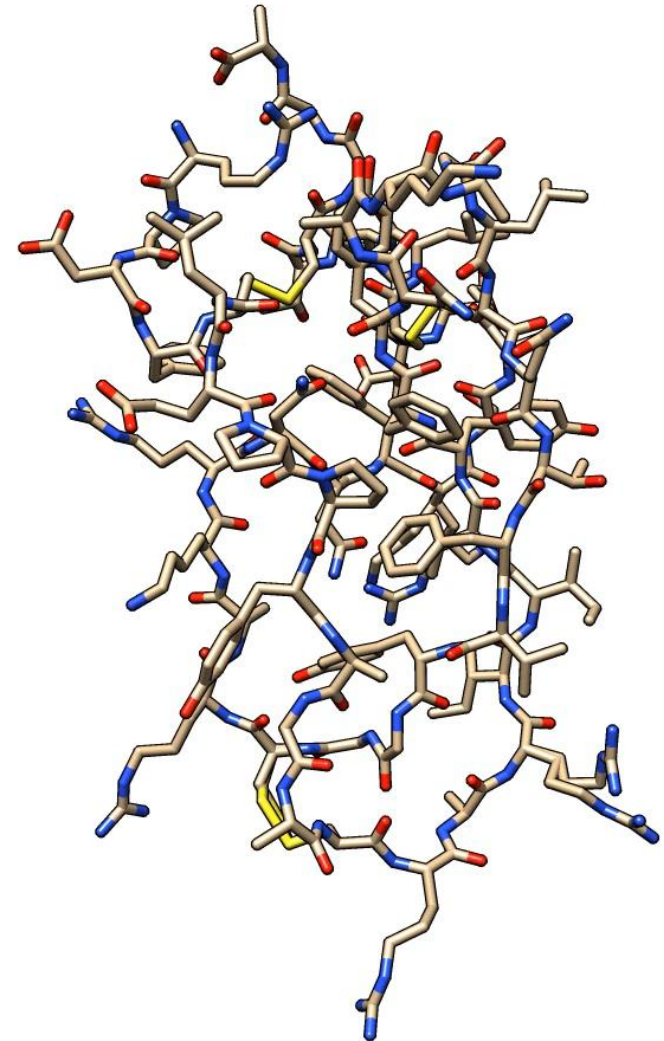
- R-free value
- Resolution
- Density coverage of ligand heavy atoms
- Active site density coverage
- Alternative locations active site/ligands
- [...]

	HT	MT	NT
Ligand	>0.9	<0.9 and >0.5	<0.5
Active Site	>0.95	<0.95 and >0.5	<0.5



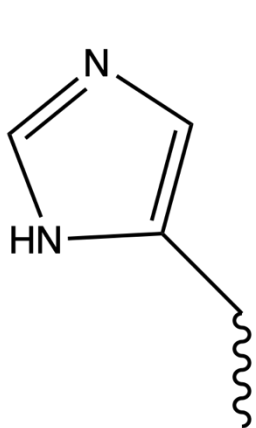
Missing information in structures: Hydrogen atoms

- Missing from most crystal structures
- Needed for molecular modelling
- Most MD packages (AMBER, CHARMM, GROMACS, etc.) include tools to “automatically” add H-atoms.
- **Any issues?**

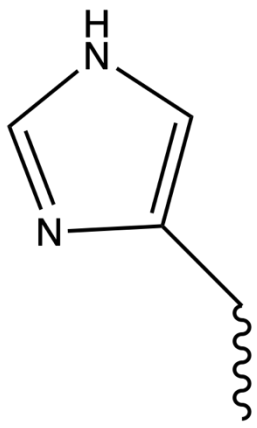


How to choose the right tautomer?

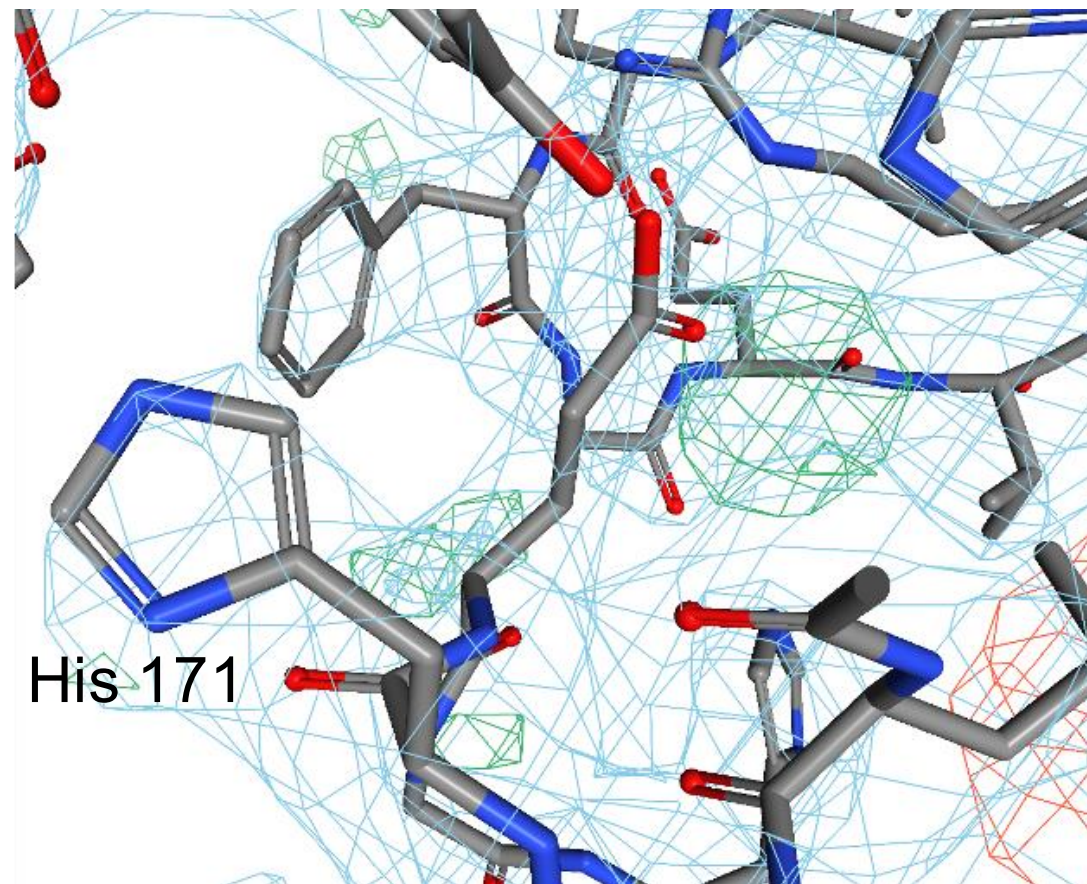
Which tautomer?



δ -tautomer

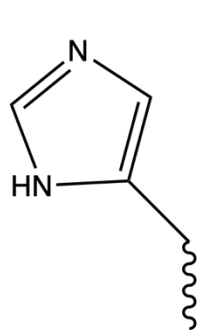


ϵ -tautomer

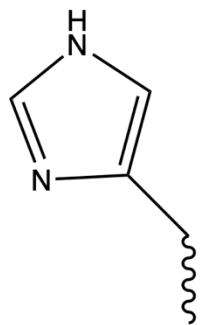


How to choose the right tautomer and protonation state?

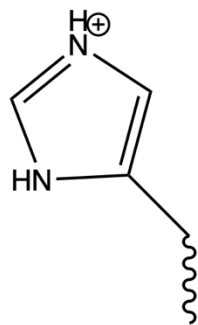
Which tautomer and protonation?



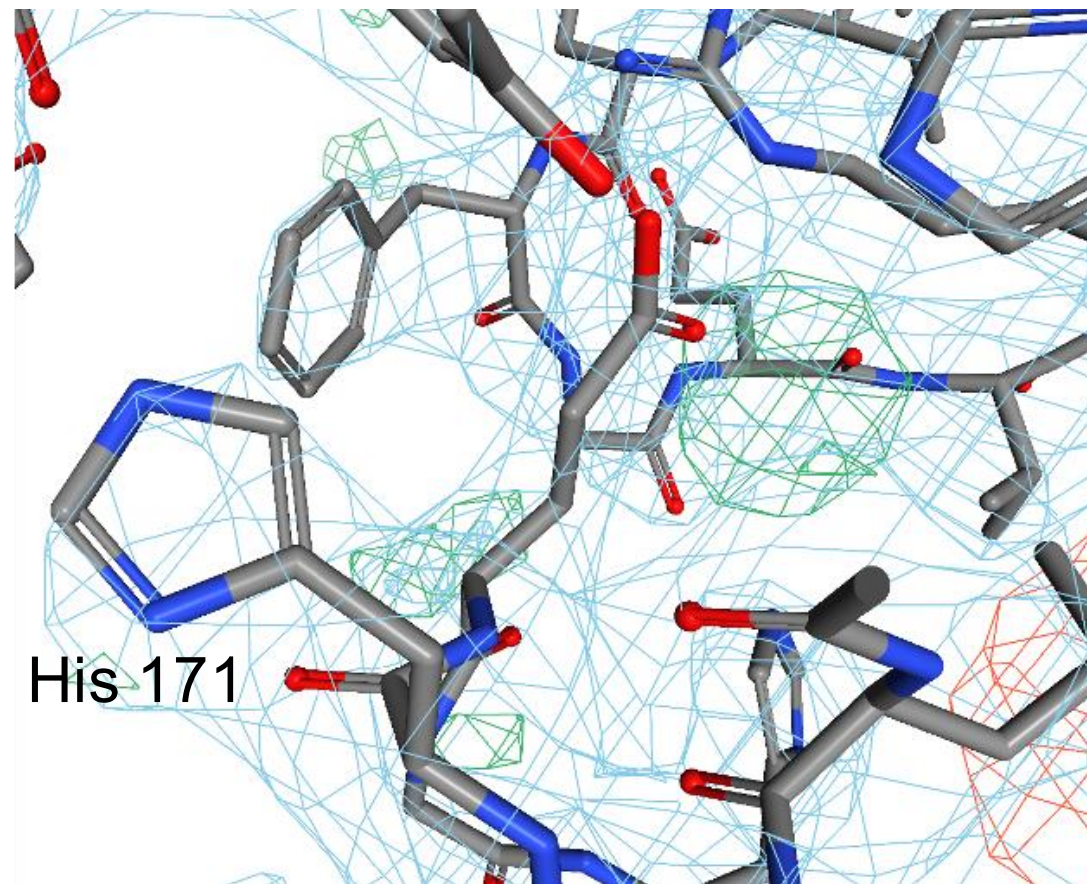
δ -tautomer



ϵ -tautomer

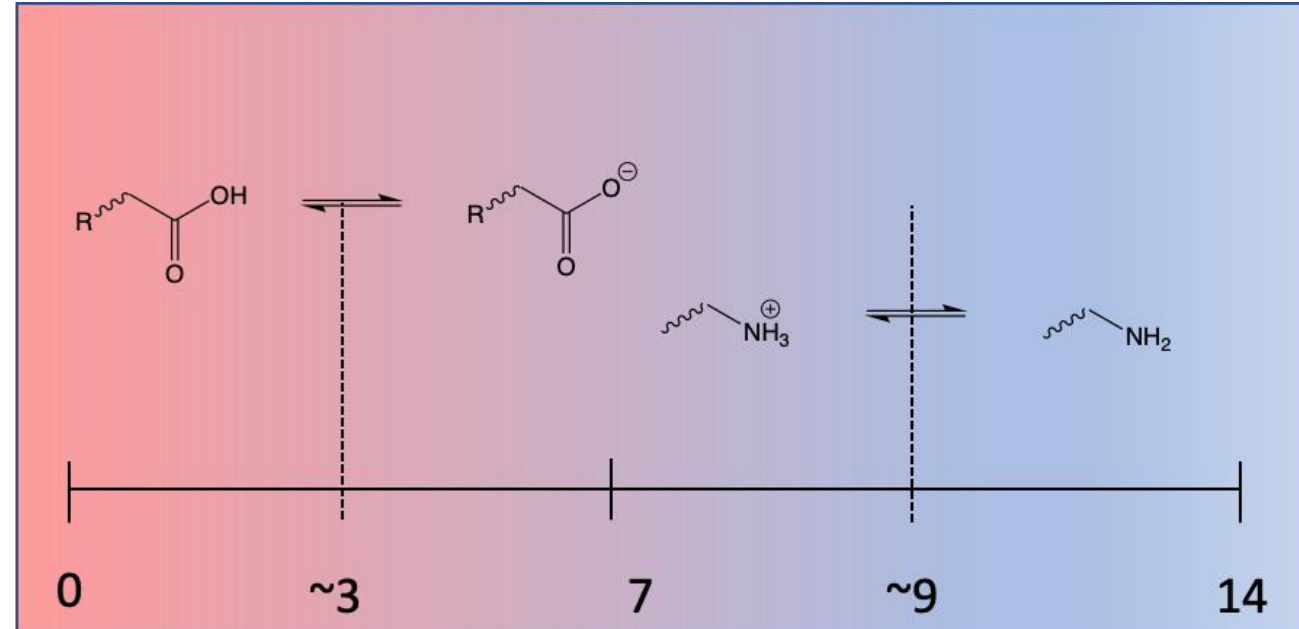


protonated



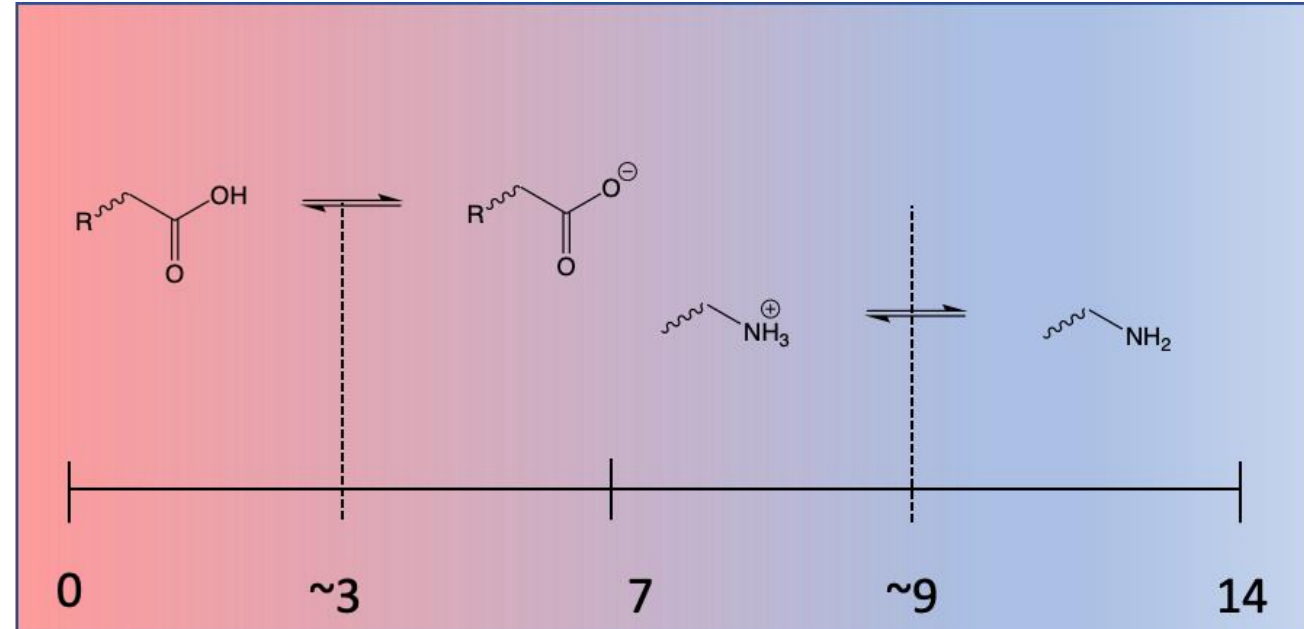
We need to worry about pKa to decide protonation

- pKa: pH at which an acidic/basic group is 50% protonated/deprotonated.
- pKas are not fixed things!
- “Standard” values refer to the situation when the group is in dilute aqueous solution.
- Groups buried in the centre of hydrophobic proteins or close to other charged groups can show large pKa shifts.



We need to worry about pKa to decide protonation

- Finding the right protonation state for proteins and ligands can be challenging!
- Poorly chosen protonation states
 - can lead to inaccurate simulations by (de)stabilising interactions
 - can lead to incorrectly estimating binding affinities or protein local or collective dynamics.

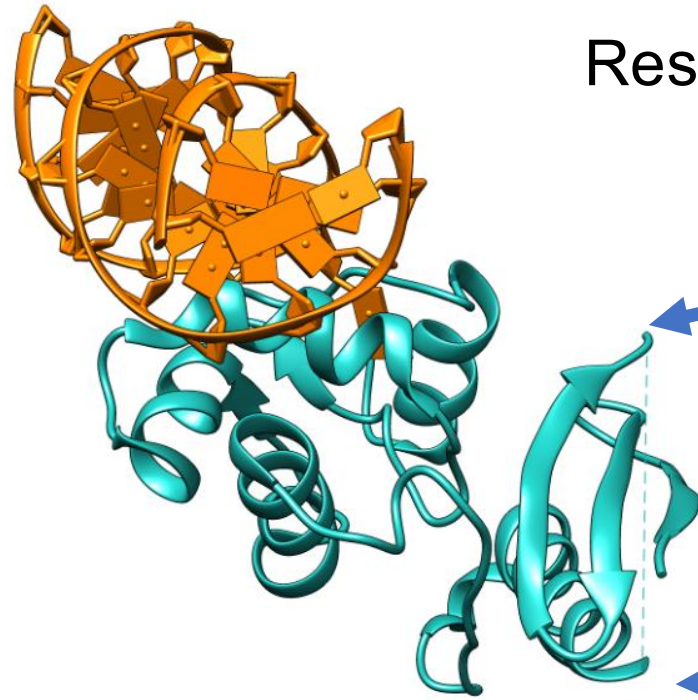


Amino acids that need protonation state consideration

Amino acid	pKa	options	Significance*
Aspartic acid	3.65	–COOH instead of COO ⁻ ?	possible
Glutamic acid	4.25	–COOH instead of COO ⁻ ?	possible
Histidine	6.00	protonated instead of neutral?	very possible
Cysteine	8.18	–S ⁻ instead of SH?	very possible
Tyrosine	10.07	–O ⁻ instead of OH?	possible
Lysine	10.53	–NH ₂ instead of NH ₃ ⁺ ?	possible
Arginine	12.48	neutral instead of protonated?	unlikely

* for a simulation around physiological pH

What if the best structure has missing residues?



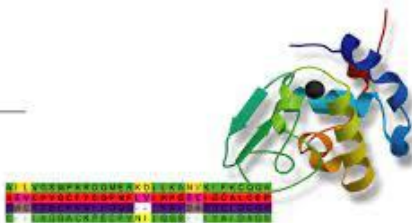
Residues 36 to 55 missing

```
REMARK 465 MISSING RESIDUES
REMARK 465 THE FOLLOWING RESIDUES WERE NOT LOCATED IN THE
REMARK 465 EXPERIMENT. (M=MODEL NUMBER; RES=RESIDUE NAME; C=CHAIN
REMARK 465 IDENTIFIER; SSSEQ=SEQUENCE NUMBER; I=INSERTION CODE.)
REMARK 465
REMARK 465  M RES C SSSEQI
```

```
REMARK 465    LYS A   36
REMARK 465    GLY A   37
REMARK 465    THR A   38
REMARK 465    SER A   39
REMARK 465    ALA A   40
REMARK 465    ALA A   41
REMARK 465    ASP A   42
REMARK 465    ALA A   43
REMARK 465    VAL A   44
REMARK 465    GLU A   45
REMARK 465    VAL A   46
REMARK 465    PRO A   47
REMARK 465    ALA A   48
REMARK 465    PRO A   49
REMARK 465    ALA A   50
REMARK 465    ALA A   51
REMARK 465    VAL A   52
REMARK 465    LEU A   53
REMARK 465    GLY A   54
REMARK 465    GLY A   55
```

Modeller

Program for Comparative Protein
Structure Modelling by Satisfaction
of Spatial Restraints



**PDB
FIXER**

AlphaFold

AlphaFold2 structures have no missing residues or atoms

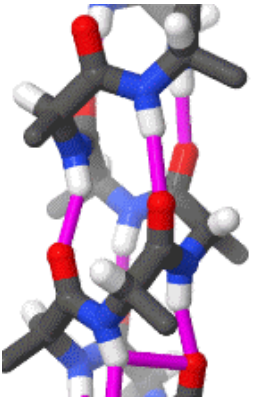
AlphaFold

G T R Y A G K V V

G T R K V V



No missing residues



Has hydrogens



AlphaFold2 structures can have unreliable areas

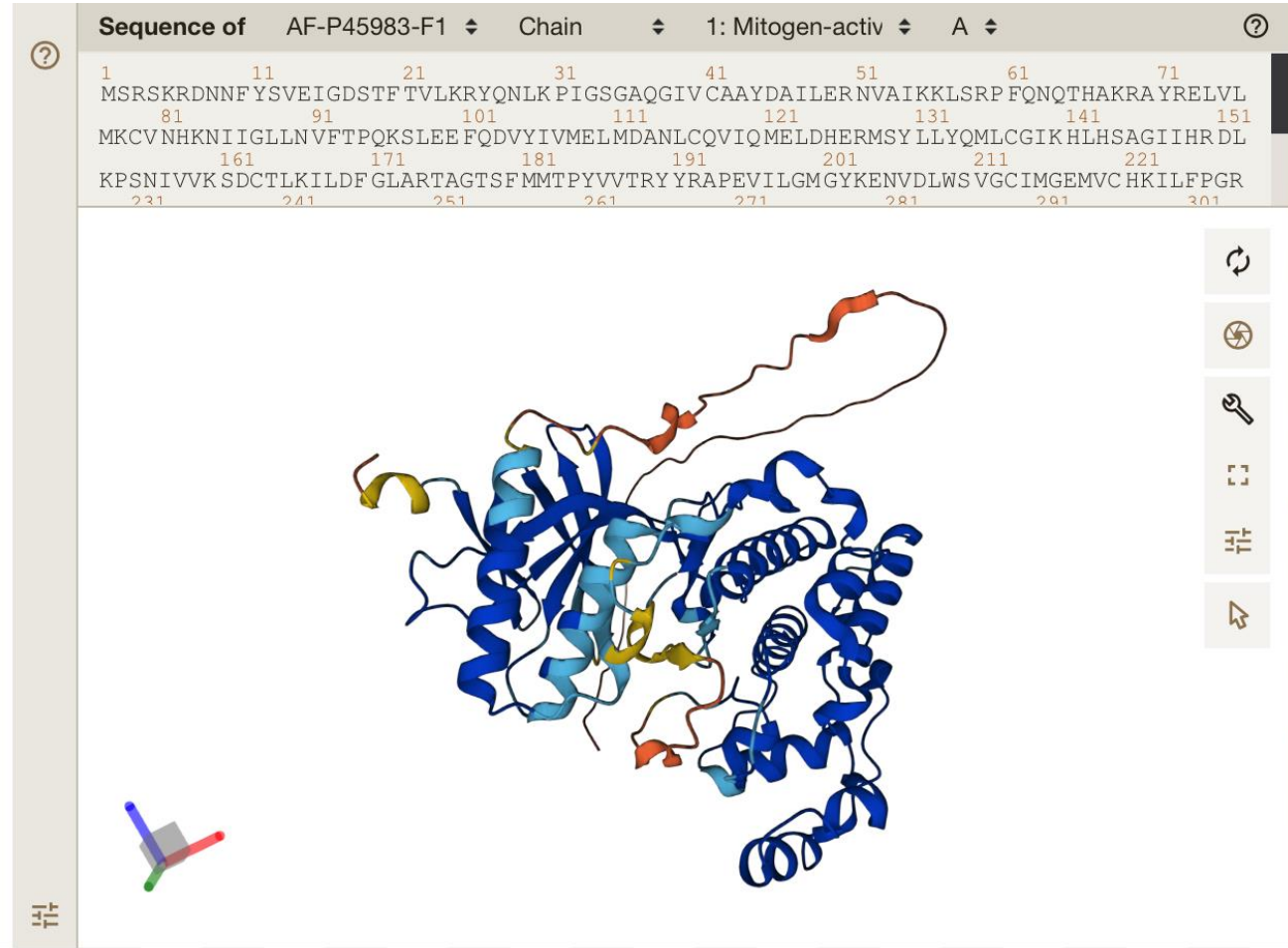
JNK1 — again

3D viewer [?](#)

Model Confidence:

- Very high (pLDDT > 90)
- Confident (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

AlphaFold produces a per-residue confidence score (pLDDT) between 0 and 100. Some regions below 50 pLDDT may be unstructured in isolation.



What is missing in this very good structure?

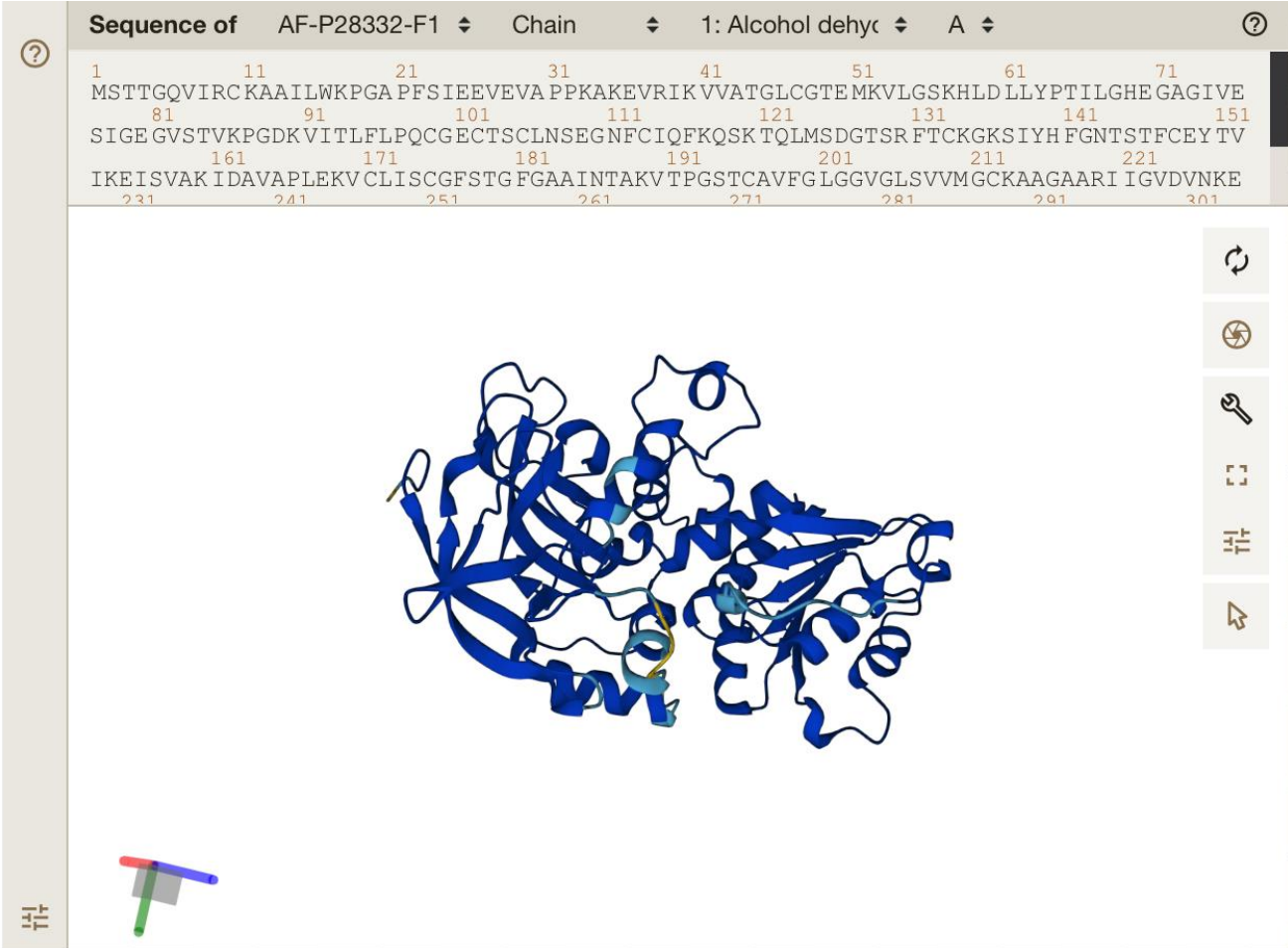
Alcohol dehydrogenase

3D viewer

Model Confidence:

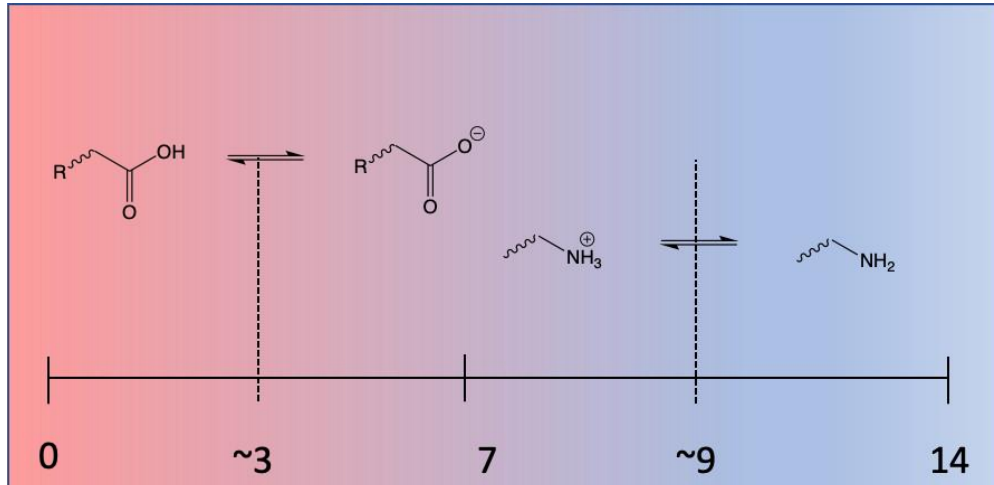
-  Very high (pLDDT > 90)
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-  Very low (pLDDT < 50)

AlphaFold produces a per-residue confidence score (pLDDT) between 0 and 100. Some regions below 50 pLDDT may be unstructured in isolation.

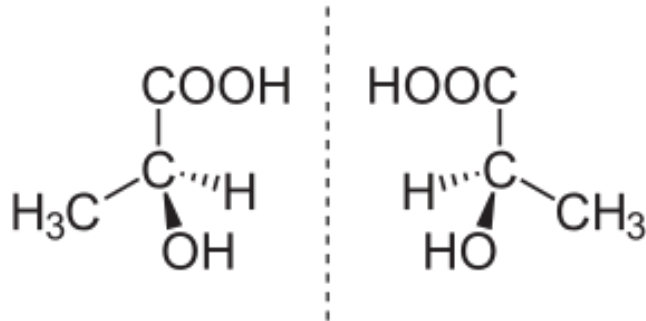


What about ligands and co-factors?

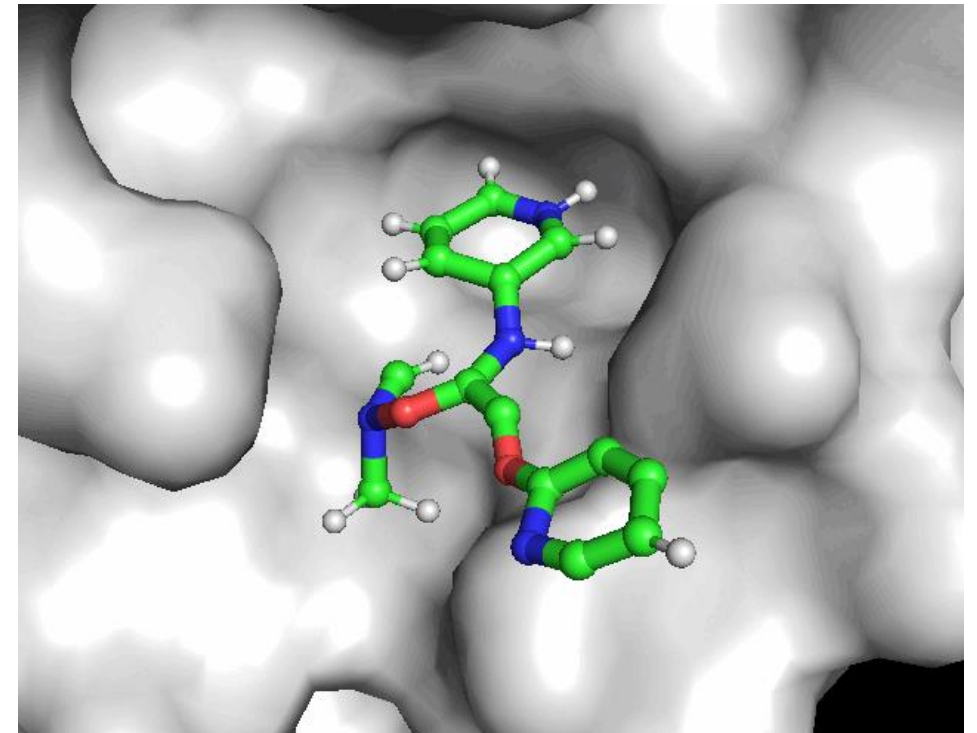
pKa for ligands is also important!



Choosing the right enantiomer



How do you get a ligand position when there is no crystal structure?



Trypsin: 1TRN

Task 1:

You have been told to simulate Trypsin starting from the PDB code *1TRN*. Have a look at the 1TRN entry in the pdb and note down what you see and choices you make. Think about:

- Stoichiometry (monomer or dimer protein?)
- PTMs (does it have post-translational modifications?)
- Non-protein molecules (e.g., cholesterol, glycol, ... simulate or remove?)
- Disulphides (does it have potential disulphide bridges?)
- Histidines (what will be the protonation state of histidines?)



https://www.rcsb.org/structure/1TRN



RCSB PDB

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Biological Assembly 1 ?



Explore in 3D: [Structure](#) | [Sequence Annotations](#)
| [Electron Density](#) | [Validation Report](#) |
[Ligand Interaction \(ISP\)](#)

Global Symmetry: Asymmetric - C1

Global Stoichiometry: Homo 2-mer - A2



1TRN

CRYSTAL STRUCTURE OF HUMAN TRYPSIN 1: UNEXPECTED PHOSPHORYLATION OF TYROSINE 151

PDB DOI: <https://doi.org/10.2242/pdb1TRN/pdb>

Classification: **HYDROLASE (SERINE PROTEINASE)**

Organism(s): [Homo sapiens](#)

Mutation(s): No

Deposited: 1995-03-16 Released: 1995-06-03

Deposition Author(s): [Gaboriaud, C.](#), [Fontecilla-Camps, J.C.](#)

Experimental Data Snapshot

Method: X-RAY DIFFRACTION

Resolution: 2.20 Å

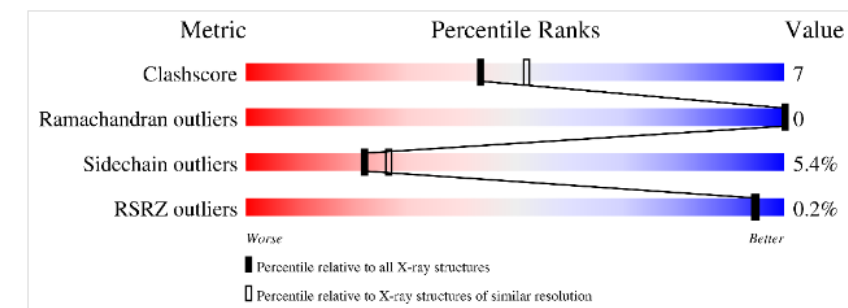
R-Value Work: 0.177

R-Value Observed: 0.177

wwPDB Validation

3D Report

Full Report

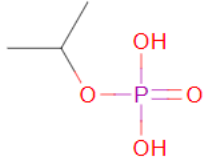


[←](#)
[↻](#)
<https://www.rcsb.org/structure/1TRN>
A
☆
📏
☆
📁
🔒

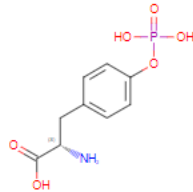
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Small Molecules

Ligands 1 Unique

ID	Chains ⓘ	Name / Formula / InChI Key	2D Diagram	3D Interactions
ISP Query on ISP Download Ideal Coordinates CCD File ⓘ Download Instance Coordinates ▾	C [auth A], D [auth B]	PHOSPHORYLISOPROPANE C ₃ H ₉ O ₄ P QPPQHRDVPBTVEV-UHFFFAOYSA-N		Interactions ▾ Interactions & Density ▾

Modified Residues ⓘ 1 Unique

ID	Chains ⓘ	Type	Formula	2D Diagram	Parent
PTR Query on PTR	A, B	L-PEPTIDE LINKING	C ₉ H ₁₂ N O ₆ P		TYR

Find proteins for [P07477](#) (*Homo sapiens*)

Explore [P07477](#) ⓘ

Go to UniProtKB: [P07477](#)

GTEX: [ENSG00000204983](#)

Entity Groups ⓘ

Sequence Clusters

[30% Identity](#) 📄

[50% Identity](#) 📄

[70% Identity](#) 📄

[90% Identity](#) 📄

[95% Identity](#) 📄

[100% Identity](#) 📄

UniProt Group

[P07477](#) 📄

Sequence Annotations

Expand

Reference Sequence

1TRN_1 ▾

0 20 40 60 80 100 120 140 160 180 200 220

1TRN_1

MODIFIED MONOMER

HYDROPATHY

DISORDER

DISORDERED BINDING

PFAM

←

↺

🔒

https://www.uniprot.org/uniprotkb/P07477/entry

🏠

A

☆

📄

🔍

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📧

📌

UniProt

BLAST

Align

Peptide search

ID mapping

SPARQL

UniProtKB ▾

Advanced | List

Search

🏠

🔖

📧

Help

Function

Names & Taxonomy

Subcellular Location

Disease & Variants

PTM/Processing

Expression

Interaction

Structure

Family & Domains

Sequence

Similar Proteins

Proteinⁱ

Geneⁱ

Statusⁱ

Organismⁱ

Serine protease 1

PRSS1

🌟 UniProtKB reviewed (Swiss-Prot)

Homo sapiens (Human)

Amino acids

Protein existenceⁱ

Annotation scoreⁱ

247 (go to sequence)

Evidence at protein level

5/5

Entry

Variant viewer 524

Feature viewer

Genomic coordinates

Publications

External links

Hi

BLAST

📄 Download

🔖 Add

Add a publication

Entry feedback

Functionⁱ

Has activity against the synthetic substrates Boc-Phe-Ser-Arg-Mec, Boc-Leu-Thr-Arg-Mec, Boc-Gln-Ala-Arg-Mec and Boc-Val-Pro-Arg-Me. The single-chain form is more active than the two-chain form against all of these substrates.

📌 1 Publication

⚠️ Caution

Tyr-154 was proposed to be phosphorylated (PubMed:8683601) but it has been shown (PubMed:17087724) to be sulfated instead. Phosphate and sulfate groups are similar in mass and size, and this can lead to erroneous interpretation of the results.

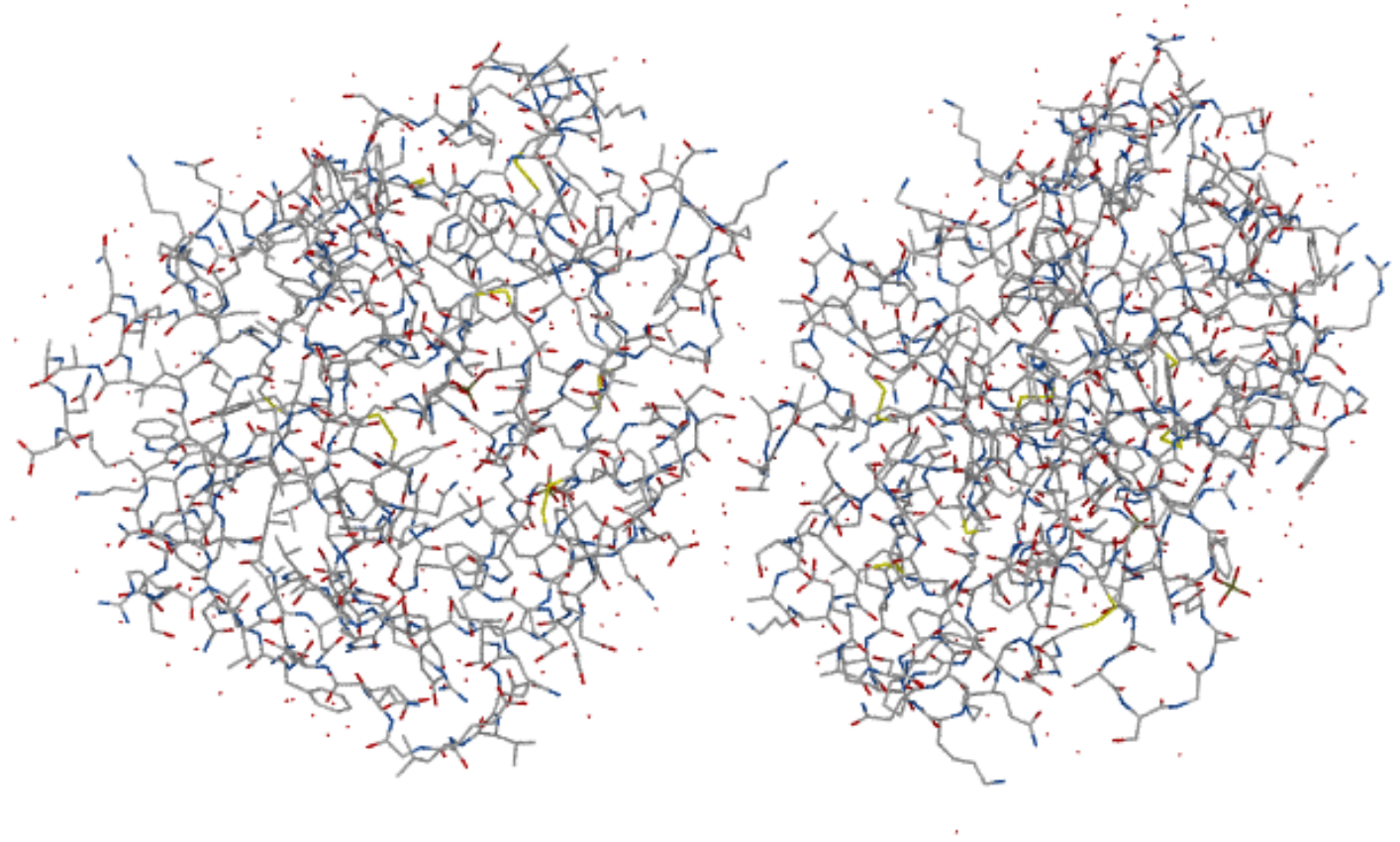
📌 2 Publications

Feedback

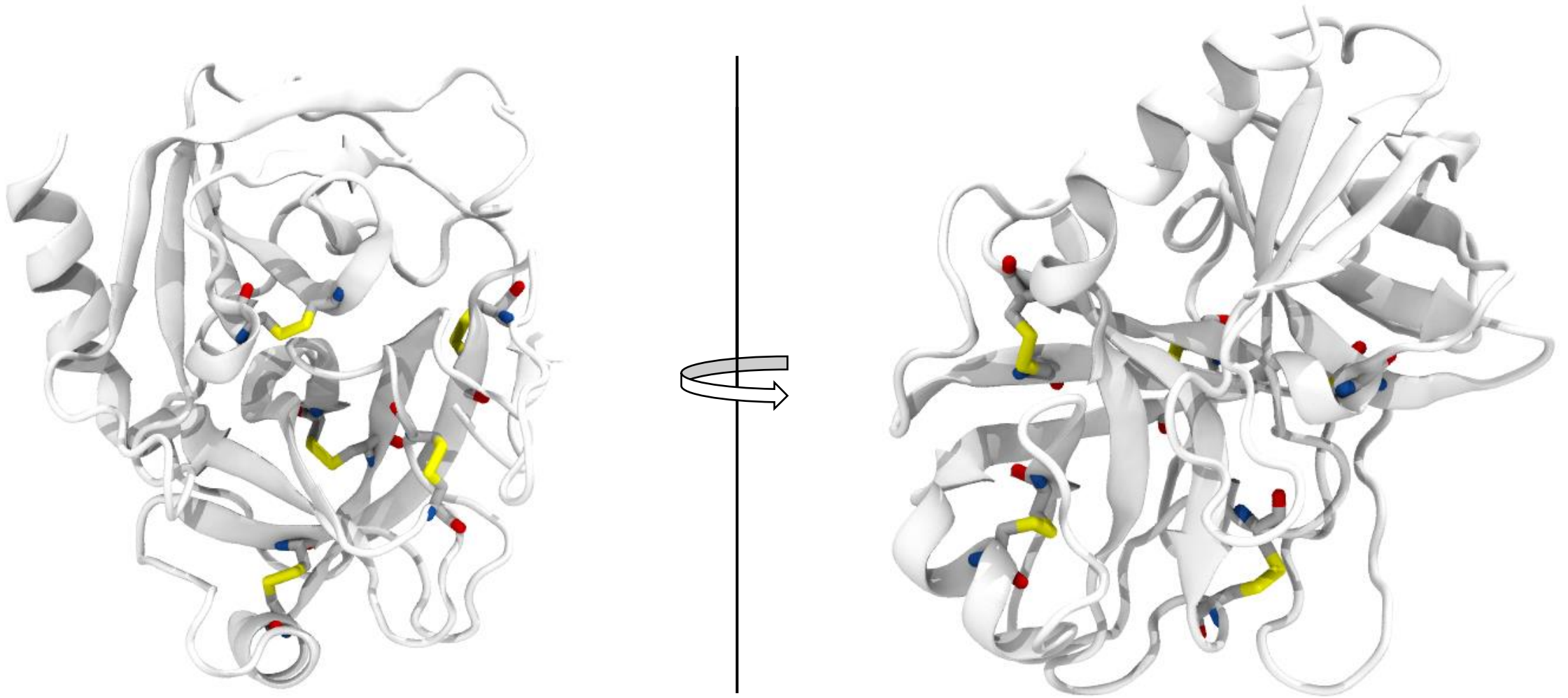
Help

27

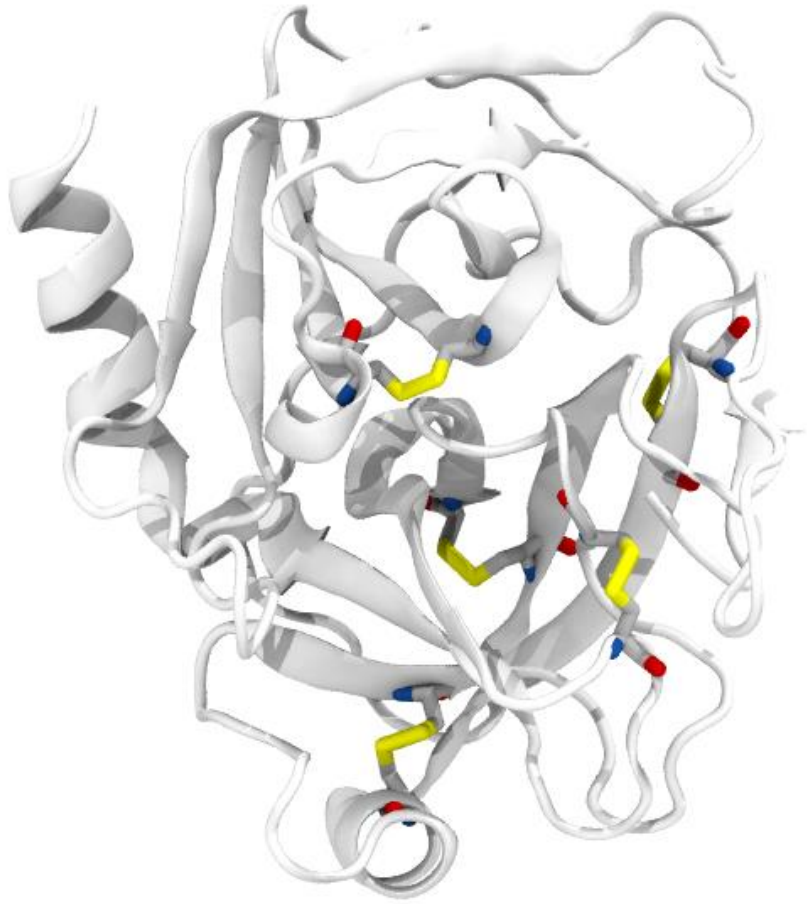
Non-protein molecules



Disulphide bridges

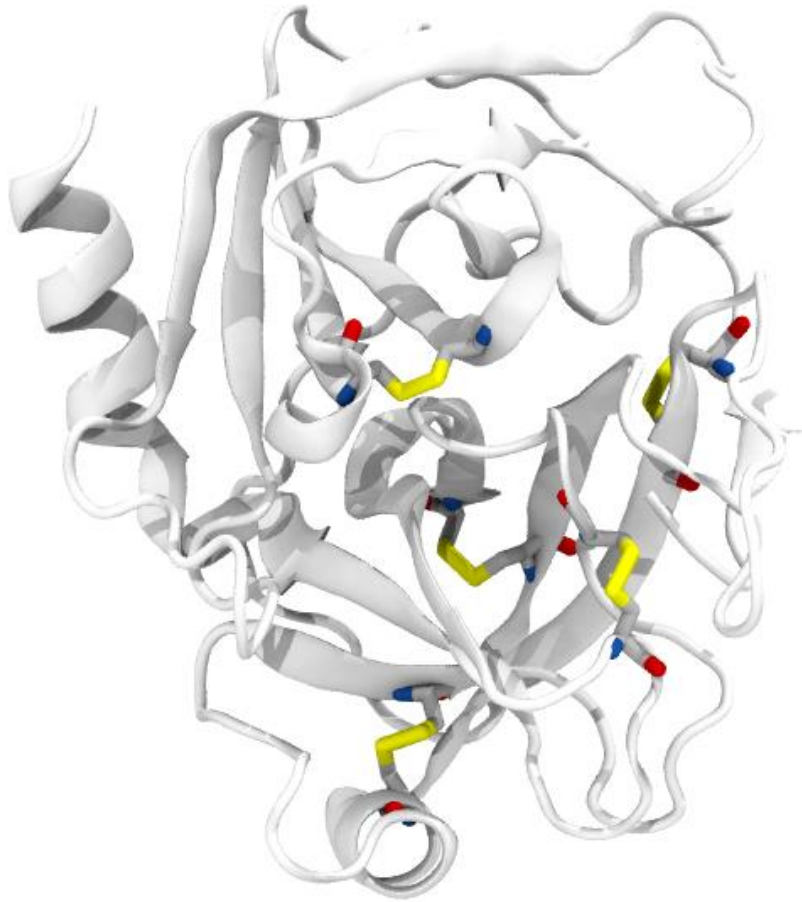


Disulphide bridges



How do you know if your protein contains disulphide bridges?

Disulphide bridges



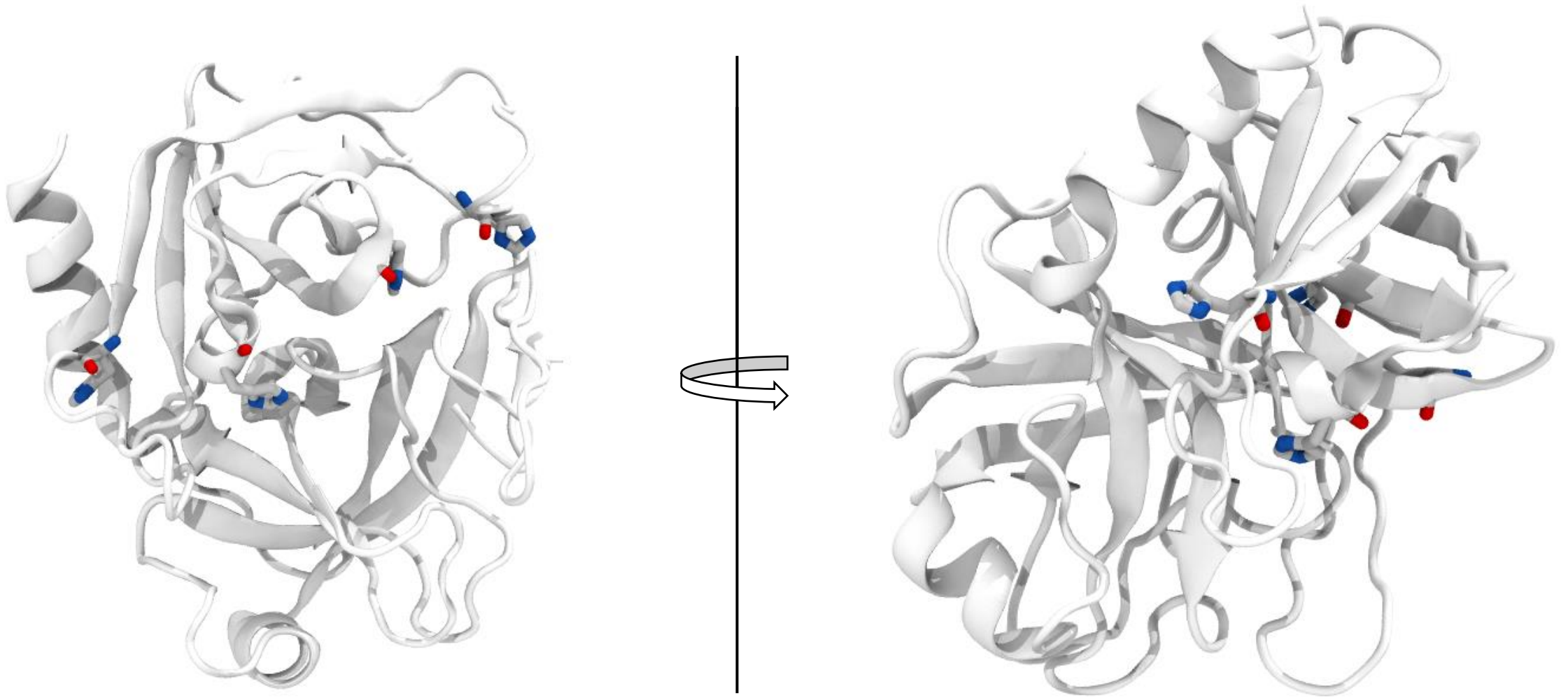
How do you know if your protein contains disulphide bridges?

Check the PDB file for SSBOND entries:

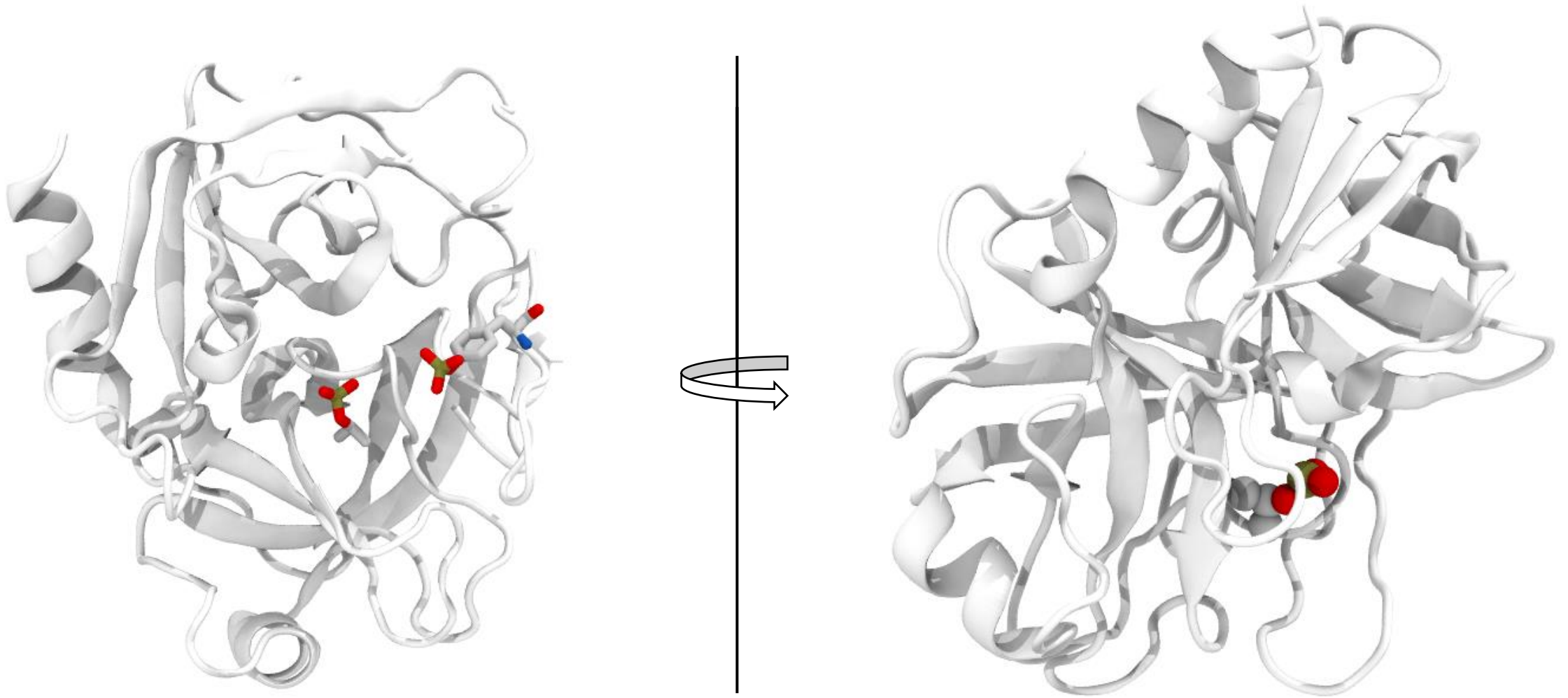
```
SHEET 7 D 7 ALA B 160 LEU B 163 -1 N LEU B 163 O CYS B 182
SSBOND 1 CYS A 22 CYS A 157 1555 1555 2.05
SSBOND 2 CYS A 42 CYS A 58 1555 1555 2.08
SSBOND 3 CYS A 136 CYS A 201 1555 1555 2.04
SSBOND 4 CYS A 168 CYS A 182 1555 1555 2.04
SSBOND 5 CYS A 191 CYS A 220 1555 1555 2.03
SSBOND 6 CYS B 22 CYS B 157 1555 1555 2.03
SSBOND 7 CYS B 42 CYS B 58 1555 1555 2.03
SSBOND 8 CYS B 136 CYS B 201 1555 1555 2.03
SSBOND 9 CYS B 168 CYS B 182 1555 1555 2.03
SSBOND 10 CYS B 191 CYS B 220 1555 1555 2.03
LINK C ASP A 150 N PTR A 151 1555 1555 1.33
```

Important! Some engines (e.g. **AMBER**) require you to manually add a bond to correctly model disulphide bridges!

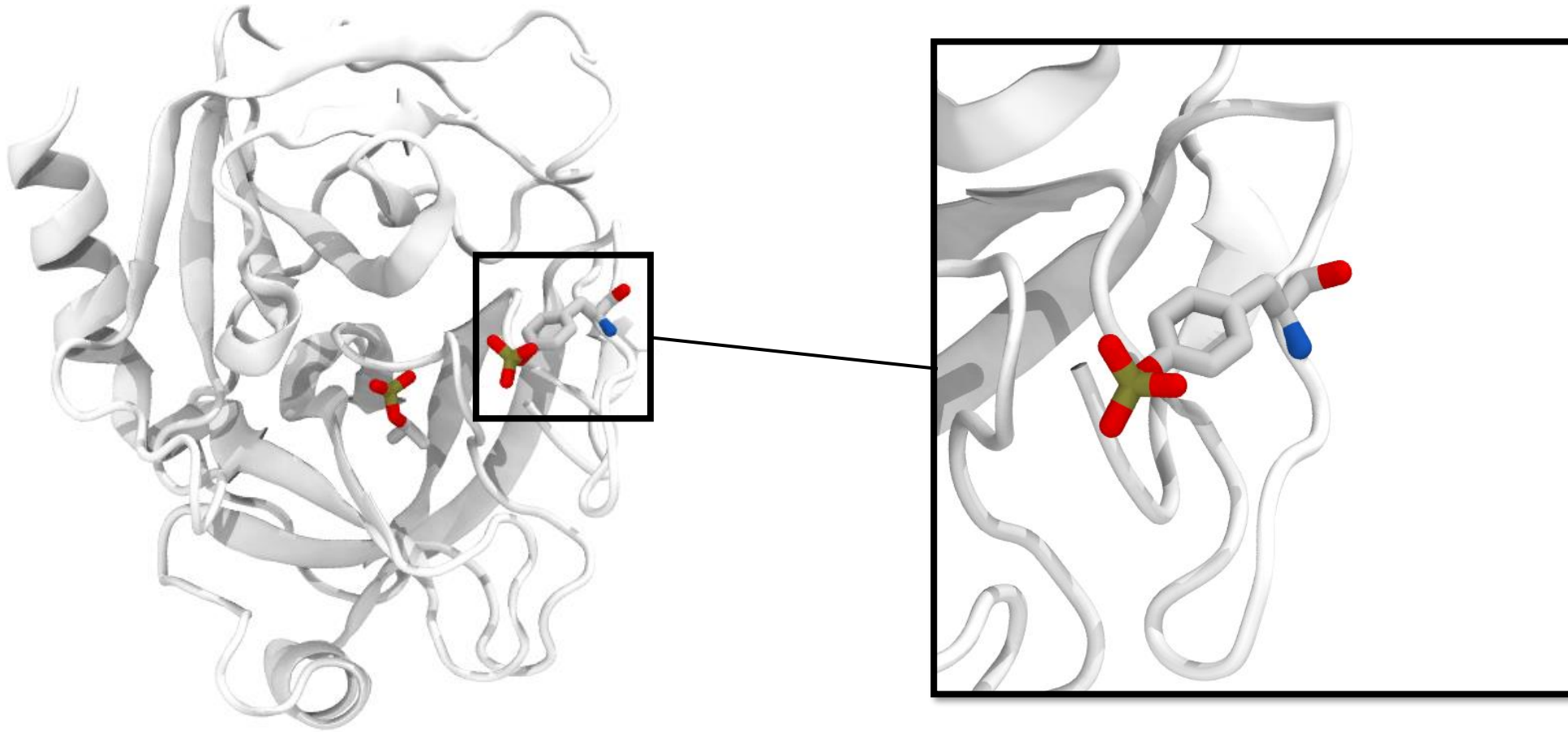
Histidines



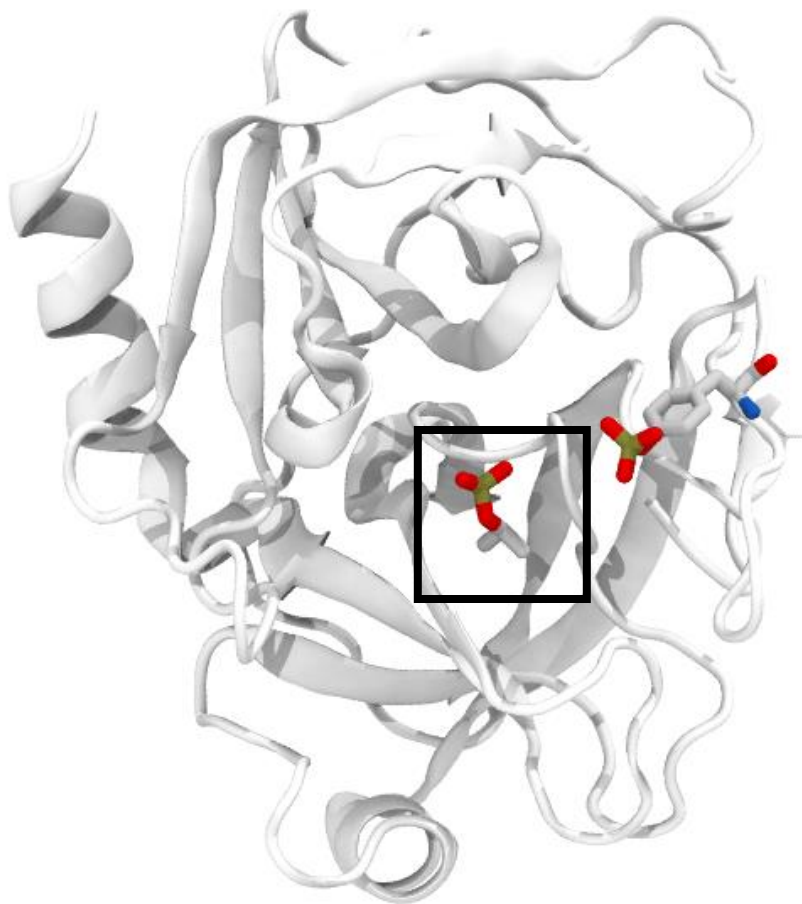
Amino acid modifications: TYR151



Amino acid modifications: TYR151



Ligand?



← ↻ 🔒 [https://en.wikipedia.org/wiki/TAPS_\(buffer\)](https://en.wikipedia.org/wiki/TAPS_(buffer)) A 📖 ☆ 📄 ☆ 📁 📌 ...

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From Wikipedia, the free encyclopedia

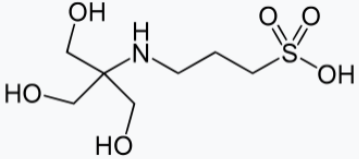
TAPS
([tris(hydroxymethyl)methylamino]propanesulfonic acid) is a chemical compound commonly used to make [buffer solutions](#).

It can bind [divalent cations](#), including [Co\(II\)](#) and [Ni\(II\)](#).^[1]

TAPS is effective to make buffer solutions in the [pH range](#) 7.7–9.1, since it has a [pK_a](#) value of 8.44 (ionic strength I = 0, 25 °C).^[2]

The pH (and pK_a at I ≠ 0) of the buffer solution changes with concentration and temperature, and this effect may be predicted e.g. using online calculators.^[3]

TAPS



Names

Preferred IUPAC name
3-[[1,3-Dihydroxy-2-(hydroxymethyl)propan-2-yl]amino]propane-1-sulfonic acid

Other names
N-Tris(hydroxymethyl)methyl-3-aminopropanesulfonic acid

Identifiers

CAS Number	29915-38-6 🔗 ✓
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Tools that will help with all of this

Most simulation packages contain tools to help add missing H-atoms. Not all work in the same way.

Where there is more than one possible answer, not all packages will **make the same decision**.

There are two independent, well-established tools designed specifically to look at these issues, and a webserver that puts them together:

Reduce

Adds hydrogens, and tests for 'NQH flips'

Download:

<http://kinemage.biochem.duke.edu/software/reduce.php>

PropKa

Predicts amino acid protonation states

Download:

<https://github.com/jensengroup/propka-3.1>

PDB2PQR server

Interface to both Reduce and proPka

Website:

<http://server.poissonboltzman.org/pdb2pqr>



...

Let's try some protein prep out!



[uniprot.org](https://www.uniprot.org)



[rcsb.org](https://www.rcsb.org)



<https://server.poissonboltzmann.org>



Task 2:

Identify a good crystal structure for **Carboxypeptidase B2** and use the Poisson Boltzmann server to produce output for AMBER.
Starting structure PDB: 3D67 use UNIPROT to find out more!

Task 3:

Look at the AlphaFold 2 structure, what do you need to watch out for here?